

DSA Decision Chart – How to Choose the Right Data Structure

1. Based on Input Size (N)

- 1 $N \leq 100 \rightarrow$ Brute Force, Recursion, Backtracking
- 2 $N \leq 1,000 \rightarrow O(N^2)$ acceptable, Arrays, Strings
- 3 $N \leq 10^4 \rightarrow O(N) / O(N \log N)$, HashMap, Heap, Sliding Window
- 4 $N \geq 10^5 \rightarrow$ Only $O(N)$ or better, Two Pointers, Bit Manipulation

2. Based on Problem Keywords

- 1 Subarray / Substring $\rightarrow$ Sliding Window
- 2 Frequency / Count $\rightarrow$ HashMap
- 3 Top K / Kth Element $\rightarrow$ Heap
- 4 Next Greater / Smaller $\rightarrow$ Monotonic Stack
- 5 Sorted Data $\rightarrow$ Binary Search / Two Pointers

3. Based on Constraints

- 1 $O(1)$ Extra Space $\rightarrow$ In-place / Two Pointers / Bit Tricks
- 2 Multiple Queries $\rightarrow$ Prefix Sum / Segment Tree
- 3 Dynamic Updates $\rightarrow$ Fenwick Tree / Segment Tree

4. Graph & Tree Indicators

- 1 Parent-Child / Hierarchy $\rightarrow$ Tree (DFS/BFS)
- 2 Dependencies / Cycles $\rightarrow$ Graph + DFS / BFS
- 3 Shortest Path $\rightarrow$ BFS / Dijkstra

5. Interview Golden Rule

- 1 If $N \geq 10^4 \rightarrow$ Never use $O(N^2)$
- 2 Explain DS choice using constraints
- 3 Always start with brute force, then optimize